# Tzu-Hsun Kao

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## Professional Summary

Tzu-Hsun is a PhD student at the University of Colorado Boulder working with Dr. Robert Marshall. He is the lead of data processing of COSMO, a 6U CubeSat that aims to measure the Earth's magnetic field by vectorized rubidium scalars. His research focuses on wave-driven particle precipitation. MS in space science, specializing in ionospheric responses to solar eclipses, volcanic eruptions, and others. Proficient in analyzing satellite data for ionospheric study and interested in magnetometer data for geomagnetic research.

## Education

**University of Colorado Boulder, CO, USA** — Aug. 2024 – Expect Dec 2027
*Ph.D. in Aerospace Engineering Sciences*

**National Central University, Taoyuan, Taiwan** — Aug. 2021 – Jun. 2023
*M.S. in Space Science and Engineering*

**National Central University, Taoyuan, Taiwan** — Aug. 2017 – Jun. 2021
*B.S. in Atmospheric Science*

## Experience

**Research Assistant** — Aug. 2024 – present
Department of Aerospace Engineering Sciences, CU Boulder

**Research Assistant** — Jun. 2023 – Aug. 2024
Center for Astronautical Physics and Engineering (CAPE), NCU

**Research Assistant** — Aug. 2021 – Jun. 2023
Department of Space Science and Engineering, NCU

**Intern** — Jul. 2023 – Aug. 2023
Taiwan Space Agency (formerly the National Space Organization)

## Awards and Honors

**2nd Place Prize**, Student Poster Competition in CEDAR Workshop, USA — Jun. 2023

**Scholarship for Excellent Students** in NCU, Taiwan — Mar. 2023

**Scholarship for Academic Outstanding Students** in NCU, Taiwan — Oct. 2021

**Scholarship for Academic Outstanding Students** in NCU, Taiwan — Oct. 2020

**Academic Achievement Award** in NCU, Taiwan — Sep. 2018

## Publications

Wu, T. Y., Liu, J. Y., Lin, C. Y., Chao, C. K., **Kao, T. H.**, an Lee, P. H. (2025). The ionospheric plasma density hole on the 10 May 2024 Mother's day great magnetic storm. *Geoscience Letters*, *52*(21), doi:10.1029/2025GL115956.

Liu, T. C., Liu, J. Y., Lee, P. H., Lin, C. Y. T., **Kao, T. H.**, and Tsai, Y. L. (2025). Uplift of the ionosphere by tropospheric and thermospheric Lamb waves induced by the 2022 Tonga volcanic eruption. *Geoscience Letters*, *12*(1), doi:10.1186/s40562-025-00419-0.

Liu, J. Y., **Kao, T. H.**, Chen, Y. C., Chum, J., Chang, W. Y., Huang, B. S., Urbář, J., Hattori, K., Zhao,

Y. X., and Chang, Y. C. (2025). Doppler Frequency shifts associated with the 18 September 2022 M6.8 Taitung Earthquake observed by local CW-HF Doppler sounding systems in Taiwan. *Journal of Geophysical Research: Space Physics*, *130*(9), doi:10.1029/2025JA033876.

Liu, J. Y., **Kao, T. H.**, Liu, T. C., Huang, B. S., Lee, P. H., Sun, Y. Y., Chen, C. H., Hattori, K., Lee, I. T., Su, C. L., Terng, C. T., and Huang, T. S. (2023) Magnetic field signatures of tropospheric and thermospheric Lamb modes triggered by the 15 January 2022 Tonga volcanic eruption, *Geophysical Research Letter*, *50*(20), doi:10.1029/2023GL105393

**Kao, T. H.**, Lee, I. T., Fong, C. J., Liu, J. Y., and Chang, M. S. (2023) Investigation of the aberrant record from bus GPS receiver onboard FORMOSAT-7/COSMIC-2 satellite constellation in low Earth orbit, *Advances in Space Research*, *72*(10), doi:10.1016/j.asr.2023.08.005

## Conference Presentations and Papers (selected)

**Kao, T. H.**, Chism, C., Kaplan, O., Ellmeier, M., Knappe, S., Hughes, J., and Marshall, R. (May. 2026). In-Orbit Calibration of Vectorized Rubidium Magnetometer onboard COSMO, EGU General Assembly 2026, Vienna, Austria. [Link]

**Kao, T. H.**, Chism, C., Ellmeier, M., Knappe, S., Hughes, J., and Marshall, R. (Dec. 2025). COSMO: A 6U CubeSat Mission for Measuring the Earth's Magnetic Field by Using Vectorized Rubidium Scalars, AGU 2025, New Orleans, LA, USA. [Link]

**Kao, T. H.**, Chism, C., Ellmeier, M., and Marshall, R. (Jun. 2025). Design, Test, Calibration, and Data Processing of the Vectorized Scalar Magnetometer Onboard COSMO, CEDAR 2025, Des Moines, IA, USA. [Link]

**Kao, T. H.** and Liu, J. Y. (Dec. 2024) Ionospheric Responses of the 8 April 2024 and 21 June 2020 Solar Eclipse Using In-Situ Satellite Measurement and the CODE GIM, 2024 American Geophysical Union, Washington, DC, USA. [Link]

**Kao, T. H.** (Sep. 2023) The space weather and single event effect on commercial off-the-shelf bus GPS receivers onboard FORMOSAT-7/COSMIC-2 constellation, 2023 Conference on Weather Analysis and Forecasting, Taipei, Taiwan. [Link]

**Kao, T. H.**, Lee, I. T., Fong, C. J., Liu, J. Y., and Chang, M. S. (Jun. 2023) Study of Single Event Effects on Commercial off-the-shelf GPS Receivers Onboard FORMOSAT-7/COSMIC-2 Constellation in Low-Earth Orbit, Coupling, Energetics and Dynamics of Atmospheric Regions Program 2023 (CEDAR 2023), San Diego, CA, USA. [Link]

**Kao, T. H.** and Liu, J. Y. (Apr. 2023) Swarm, FORMOSAT-7/COSMIC-2, and ICON ionospheric observations during the annular solar eclipse on 21 June 2020, EGU General Assembly 2023, Vienna, Austria. [Link]

**Kao, T. H.** and Liu, J. Y. (Jun. 2022) Multiple Satellite Observations on the Ionosphere Response to the 21 June 2020 Annular Solar Eclipse, Taiwan Geoscience Assembly (TGA), Taipei, Taiwan.

**Kao, T. H.**, Lee, I. T., and Fong, C. J. (Nov. 2021) Aberrant Record on the Spacecraft Bus GPS Receiver on FORMOSAT-7 Mission, International Conference on Astronautics and Space Exploration in 2021 (iCASE 2021), Hsinchu, Taiwan.